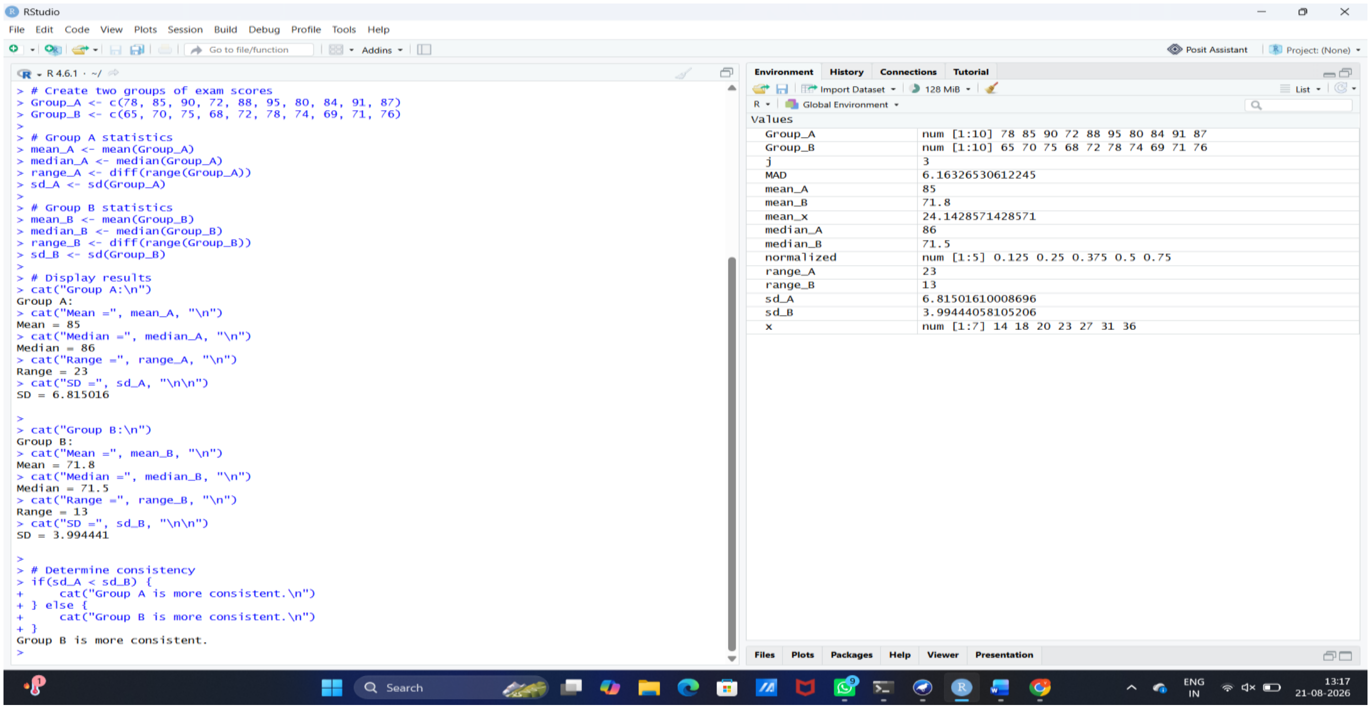
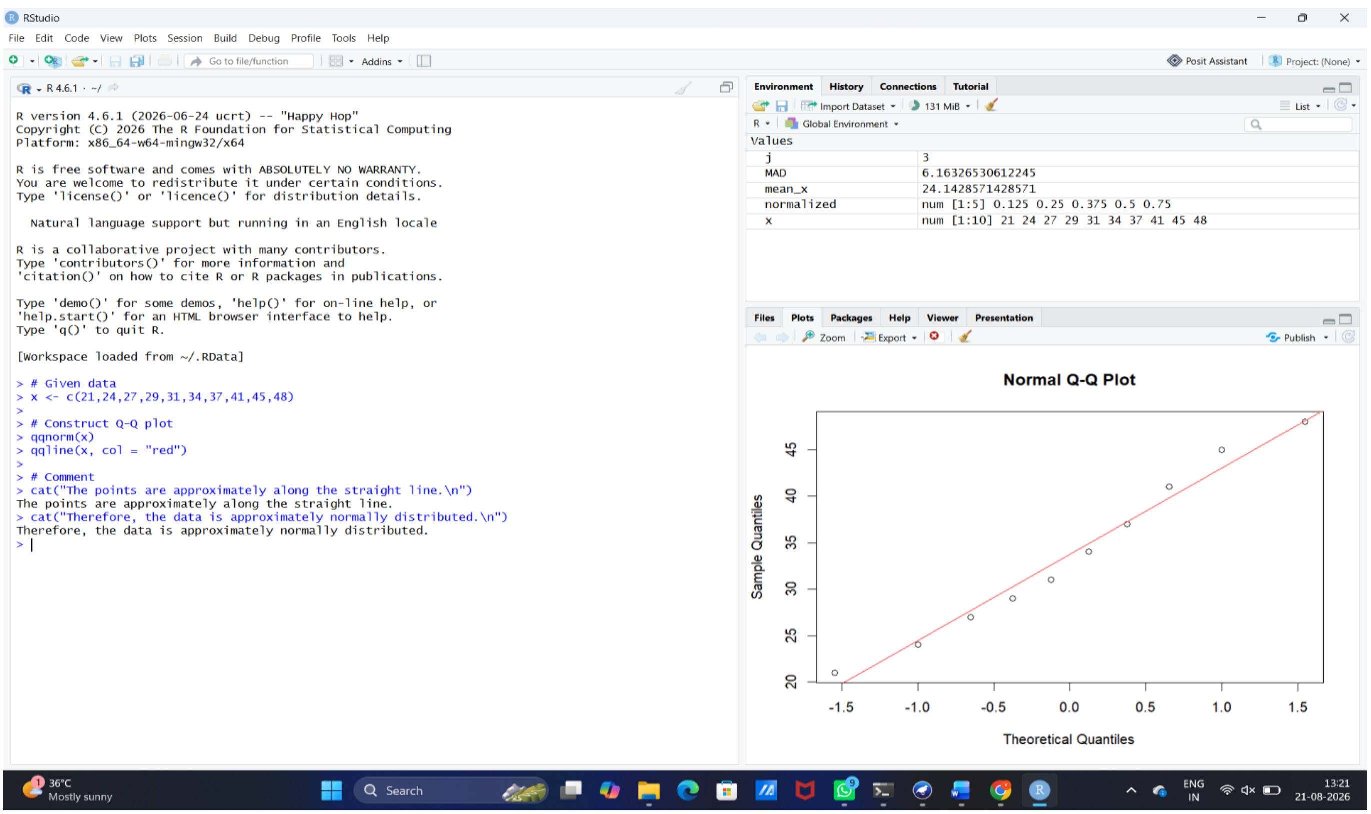
SET 6

1. R: Create two groups of exam scores; calculate group-wise mean, median, range and SD and determine consistency.



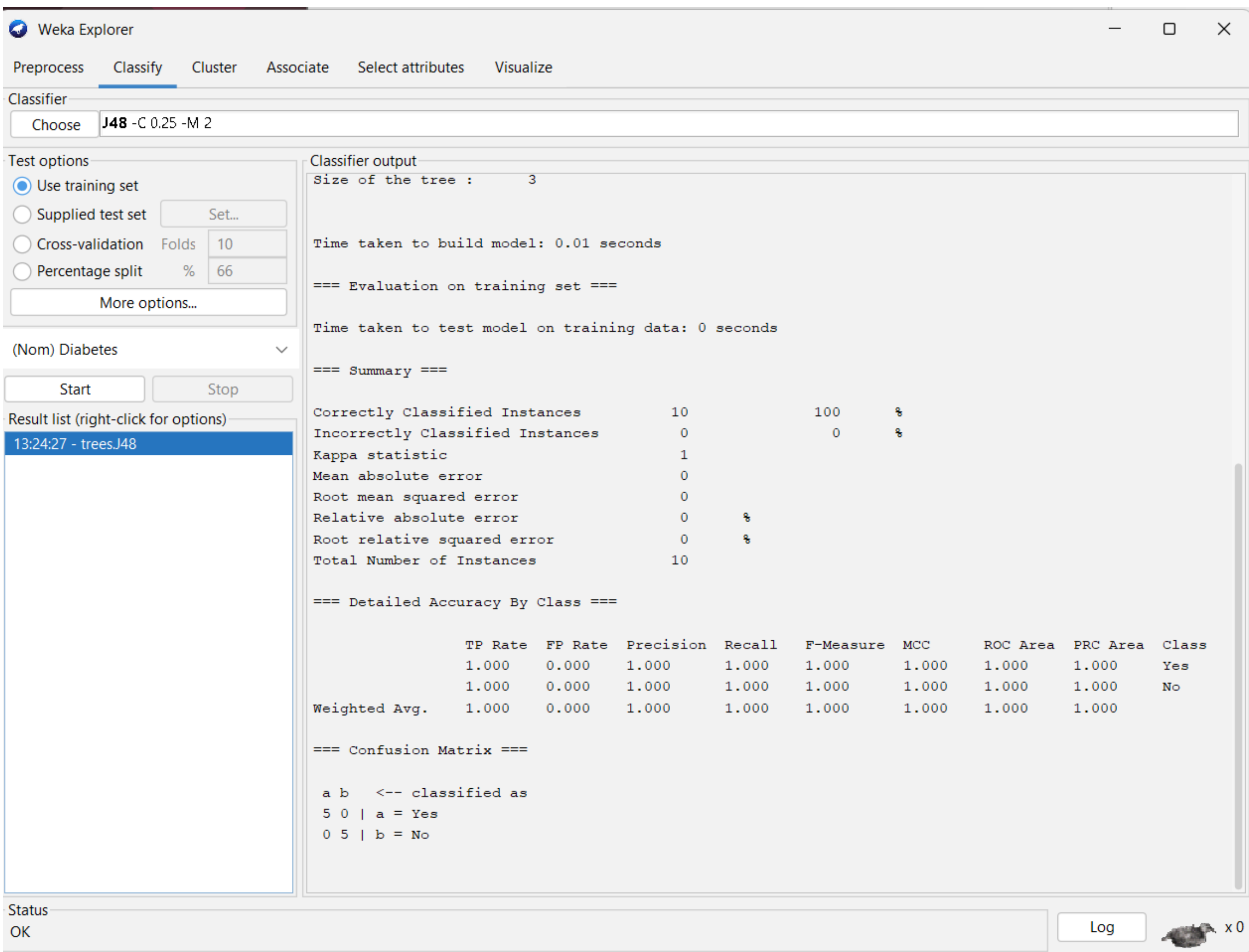
2. R: Construct a Q-Q plot for 21,24,27,29,31,34,37,41,45,48 and comment on normality.



3. WEKA: Create patient Age, Glucose, BMI, Diabetes ARFF data; apply J48 and obtain confusion matrix.

Dataset: Diabetes

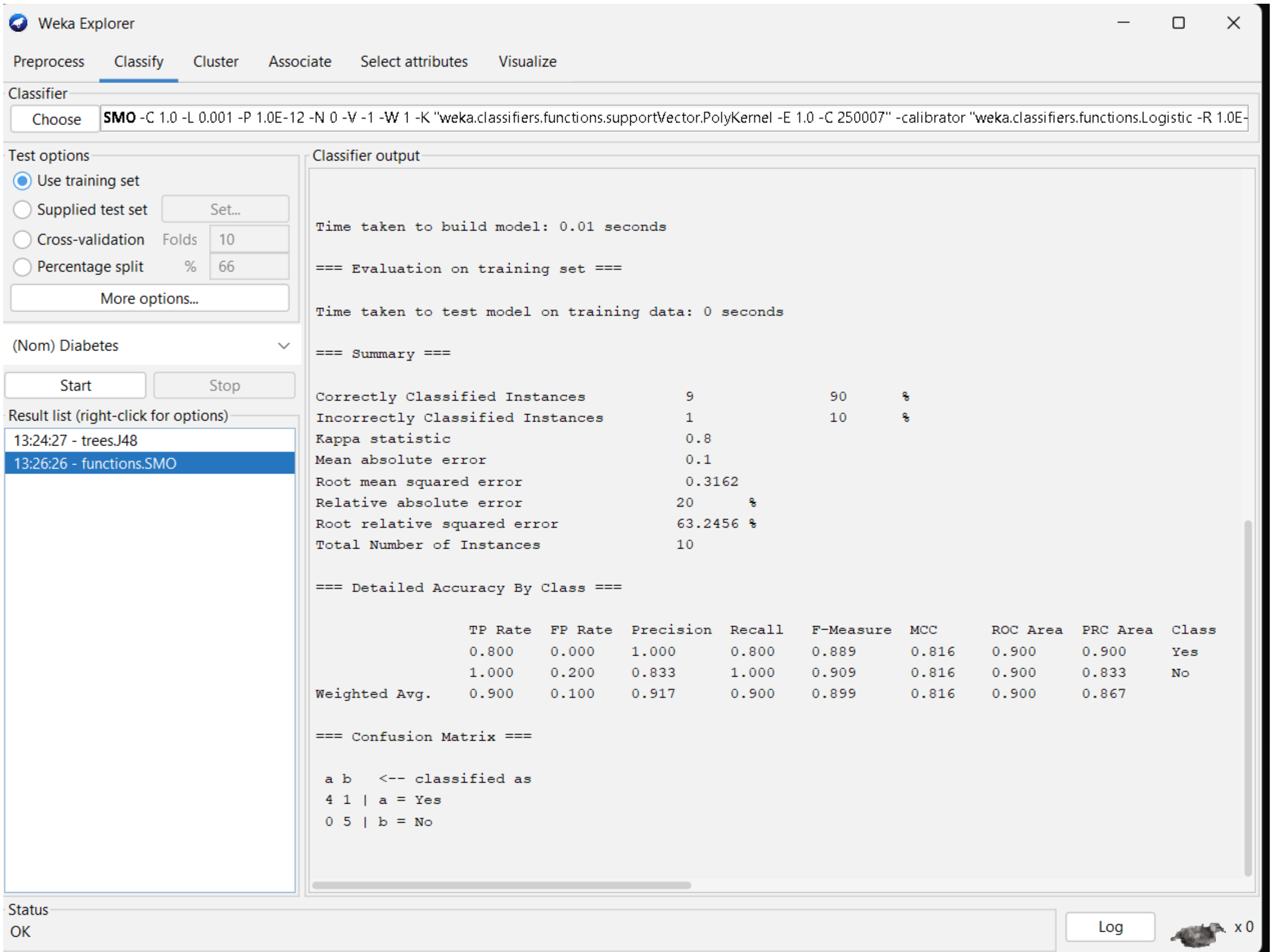
```
@relation diabetes
@attribute Age numeric
@attribute Glucose numeric
@attribute BMI numeric
@attribute Diabetes {Yes,No}
@data
25,95,22.5,No
32,130,28.1,Yes
41,145,31.2,Yes
29,105,24.8,No
50,160,34.5,Yes
36,120,27.0,No
45,150,32.1,Yes
27,100,23.5,No
55,170,36.0,Yes
39,115,26.5,No
```



4. WEKA: Apply SMO to the diabetes data and calculate accuracy, precision, recall and F1; compare with J48.

Dataset: Diabetes – SMO

Use the same Diabetes ARFF dataset from Set 6 Q3.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Diabetes
Start
Stop
Result list (right-click for options)
13:24:27 - trees.J48
13:26:26 - functions.SMO
Classifier output
Time taken to build model: 0.01 seconds
=== Evaluation on training set ===
Time taken to test model on training data: 0 seconds
=== Summary ===
Correctly Classified Instances 9 90 %
Incorrectly Classified Instances 1 10 %
Kappa statistic 0.8
Mean absolute error 0.1
Root mean squared error 0.3162
Relative absolute error 20 %
Root relative squared error 63.2456 %
Total Number of Instances 10
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.800 0.000 1.000 0.800 0.889 0.816 0.900 0.900 Yes
1.000 0.200 0.833 1.000 0.909 0.816 0.900 0.833 No
Weighted Avg. 0.900 0.100 0.917 0.900 0.899 0.816 0.900 0.867
=== Confusion Matrix ===
a b <-- classified as
4 1 | a = Yes
0 5 | b = No
Status
OK
Log
x 0